

# **X-Ray Diffraction Analysis of Paper Samples**

**EFFECTS OF WATER, DEACIDIFICATION, AND  
AGEING ON CELLULOSE CRYSTALLINITY**

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Date: 13/09/2024

# Introduction



## Research Focus:

Investigate how washing, deacidification, and artificial ageing affect the nano structural properties of cellulose in paper samples.



## Techniques Used:

X-Ray Diffraction (XRD) to study cellulose crystallinity.



## Importance:

Crystallinity is very important in the aging and preservation of historical paper materials.

# Methods and Samples



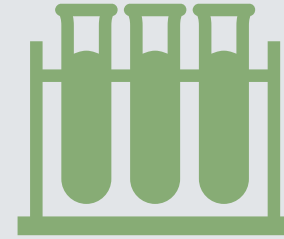
## Methods:

### 1. XRD:

Used to analyze crystallinity changes (Crystallinity Index (CI) & Crystallinity Ratio (CR)).

$$CI = \frac{I_{max} - I_{min}}{I_{max}}$$

$$CR = 1 - \frac{I_{min}}{I_{max} - I_{min}}$$



## Samples:

- Old "Rag" Paper: From 1831, mostly pure flax fibers.
- New Whatman Paper: Made from pure cotton fibers.
- Treatments:
  - **Washed in deionized water.**
  - **Deacidification with  $\text{Ca}(\text{OH})_2$  and  $\text{Mg}(\text{HCO}_3)_2$ .**
  - **Artificial aging at 80°C and 65% RH for 2 weeks.**

# Results

| Old Paper samples                               | CI    | CR    |
|-------------------------------------------------|-------|-------|
| Reference (untreated)                           | 0.660 | 0.484 |
| Reference (untreated)                           | 0.668 | 0.504 |
| Washed in deionised H <sub>2</sub> O            | 0.669 | 0.506 |
| Washed in deionised H <sub>2</sub> O            | 0.672 | 0.512 |
| Treated with Ca(OH) <sub>2</sub>                | 0.667 | 0.502 |
| Treated with Ca(OH) <sub>2</sub>                | 0.689 | 0.549 |
| Treated with Mg(HCO <sub>3</sub> ) <sub>2</sub> | 0.586 | 0.293 |
| Treated with Mg(HCO <sub>3</sub> ) <sub>2</sub> | 0.651 | 0.463 |

Table 1. Crystallinity Index and Crystallinity Ratio of the old paper samples. Crystallinity Index Std dev  $\pm$  0.0109, Crystallinity Ratio Std dev  $\pm$  0.0241

| New Paper samples                          | CI    | CR    |
|--------------------------------------------|-------|-------|
| Reference (untreated)                      | 0.725 | 0.621 |
| Reference (untreated)                      | 0.721 | 0.612 |
| Treated with Ca(OH) <sub>2</sub>           | 0.732 | 0.634 |
| Treated with Ca(OH) <sub>2</sub>           | 0.729 | 0.629 |
| Treated Mg(HCO <sub>3</sub> ) <sub>2</sub> | 0.723 | 0.618 |
| Treated Mg(HCO <sub>3</sub> ) <sub>2</sub> | 0.720 | 0.611 |

Table 2. Crystallinity Index and Crystallinity Ratio of the new paper samples. Crystallinity Index Std dev  $\pm$  0.0049, Crystallinity Ratio Std dev  $\pm$  0.0092

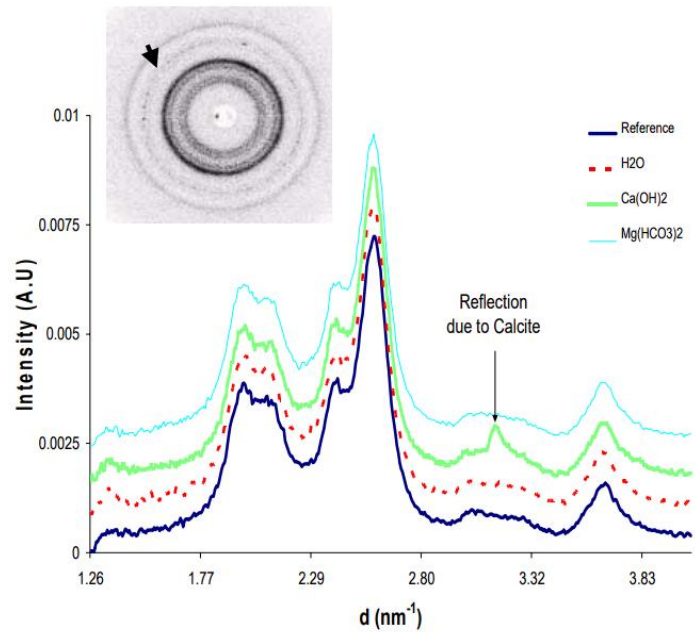


Figure 2. On the top X-ray diffraction image of old paper treated with Ca(OH)<sub>2</sub>. Calcite reflection has been labelled with an arrow. 1D linear trace of the 2D X-ray diffraction images for the old paper samples set. Calcite reflection due to treatment with Ca(OH)<sub>2</sub> has been labelled. (Linear traces offset).

| Aged Old Paper samples                          | CI    | CR    |
|-------------------------------------------------|-------|-------|
| Reference (aged)                                | 0.721 | 0.613 |
| Reference (aged)                                | 0.714 | 0.600 |
| Washed in deionised H <sub>2</sub> O            | 0.671 | 0.510 |
| Washed in deionised H <sub>2</sub> O            | 0.676 | 0.522 |
| Treated with Ca(OH) <sub>2</sub>                | 0.671 | 0.510 |
| Treated with Ca(OH) <sub>2</sub>                | 0.670 | 0.507 |
| Treated with Mg(HCO <sub>3</sub> ) <sub>2</sub> | 0.704 | 0.580 |
| Treated with Mg(HCO <sub>3</sub> ) <sub>2</sub> | 0.670 | 0.507 |

Table 3. Crystallinity Index and Crystallinity Ratio of the artificially aged old paper samples. Crystallinity Index Std dev  $\pm$  0.0220, Crystallinity Ratio Std dev  $\pm$  0.0458.

| Aged New Paper samples                          | CI    | CR    |
|-------------------------------------------------|-------|-------|
| Reference (aged)                                | 0.744 | 0.655 |
| Reference (aged)                                | 0.742 | 0.652 |
| Treated with Ca(OH) <sub>2</sub>                | 0.732 | 0.635 |
| Treated with Ca(OH) <sub>2</sub>                | 0.751 | 0.668 |
| Treated with Mg(HCO <sub>3</sub> ) <sub>2</sub> | 0.742 | 0.652 |
| Treated with Mg(HCO <sub>3</sub> ) <sub>2</sub> | 0.735 | 0.639 |

Table 4. Crystallinity Index and Crystallinity Ratio of the artificially aged new paper samples. Crystallinity Index Std dev  $\pm$  0.0066, Crystallinity Ratio Std dev  $\pm$  0.0120

# Conclusion

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**Ageing Effects:** Artificial aging induces crystallinity in cellulose fibers.

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**Conservation Impact:** Deacidification treatments do not significantly alter crystallinity in new paper, but aged paper is affected.

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**Practical Implication:** Washing and deacidification treatments help preserve the structural integrity of historical paper, particularly in preventing excessive crystallization.

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## **Future Work:**

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Investigate other complex paper types (e.g., wood fibers, lignin-containing papers) to fully understand the broader impacts of conservation treatments.

*Thank you*

